

Electroencephalogram data processing: intrinsic motivation and decision-making in patients with Parkinson's syndrome

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Abbreviations

α	Alpha (frequency band)	Ms	Millisecondes
β	Bêta (frequency band)	NaN	Not a Number
γ	Gamma (frequency band)	PD	Peak Deceleration
BEM	Boundary Element Method	PSD	Power Spectral Density
CEIC	Clinical Research Ethics	PT	Punch-Trough
Committee		PV	Peak Velocit
Coreg	Coregistration	RFE	Recursive Feature
DT	Deceleration Time	Elimination	
EEG	Electroencephalography	RF	Random Forest
FIG	Figure	ROIs	Regions of Interest
GSN129	Geodesic Sensor Net 129	SI	Stop-In
ICA	Independent Component	SRC	Source space
Analysis		TMS	Transcranial Magnetic
ITI	Inter-Trial Interval	Stimulation	
Hz	Hertz	tDCS	transcranial Direct Current
kNN	k-Nearest Neighbors	Stimulation	
MLR	Multiple Linear Regression	TTPV	Time To Peak Velocity
MDS	Movement Disorder Society		

Summary

Can intrinsic motivation, a central driver of human action, modulate brain activity even in pathological contexts? This study looks into how a motor decision task, shaped by social hierarchy, affects a patient with Parkinson's disease. We use high-density electroencephalography (EEG) recordings to check out the brain activity linked to various motivational states. Spectral and functional connectivity analyses in both electrode and source space reveal significant modulation of beta and gamma frequency bands depending on the experimental condition. Standard classification models enable accurate decoding of these states, reaching up to 90% accuracy. In practice, we extract biomarkers for the brain activity and importantly, identify cortical networks involved in social self-evaluation that modulates volitional efforts of subjects performing tasks with or without other participants. The technical part of this projects is the development of an automated pipeline based on source level reconstruction for EEG signals based on the python library MNE-Python. This work confirms the applicability of this framework to Parkinson patients, as was found earlier on healthy subjects. This opens the perspectives for a quantitative analysis of the alteration of self-evaluation processes in Parkinson patients, later aiming at developing therapeutic strategies based on cognitive and social engagement for this neurodegenerative diseases.

Résumé

La motivation intrinsèque, moteur central de l'action humaine, peut-elle moduler l'activité cérébrale même dans des contextes pathologiques ? Cette étude explore l'impact d'une tâche de décision motrice, influencée par la hiérarchie sociale, sur un patient atteint de la maladie de Parkinson. Nous utilisons des enregistrements électroencéphalographiques haute densité (EEG) pour examiner l'activité cérébrale associée à différents états motivationnels. Des analyses spectrales et de connectivité fonctionnelle, menées à la fois dans l'espace électrode et dans l'espace source, révèlent une modulation significative des bandes de fréquence bêta et gamma selon les conditions expérimentales. Des modèles de classification standards permettent de décoder avec précision ces états, atteignant jusqu'à 90 % de précision. Concrètement, nous extrayons des biomarqueurs de l'activité cérébrale et surtout, identifions les réseaux corticaux impliqués dans l'autoévaluation sociale qui module l'effort volontaire des sujets accomplissant des tâches seuls ou en présence d'autres participants. L'aspect technique de ce projet repose sur le développement d'un pipeline automatisé basé sur la reconstruction en espace source des signaux EEG, en utilisant la bibliothèque Python MNE-Python. Ce travail confirme l'applicabilité de ce cadre méthodologique aux patients parkinsoniens, comme cela avait été démontré précédemment chez les sujets sains. Cela ouvre la voie à une analyse quantitative de l'altération des processus d'autoévaluation chez les patients atteints de Parkinson, avec pour objectif ultérieur de développer des stratégies thérapeutiques fondées sur l'engagement cognitif et social dans cette maladie neurodégénérative.

Introduction

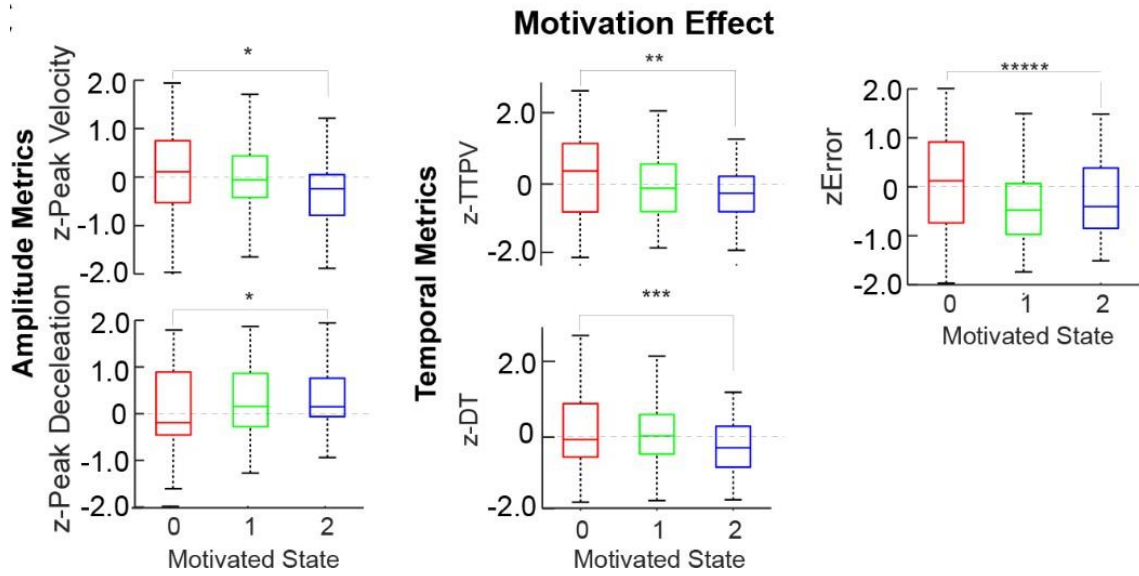
Motivation plays a central role in how we behave and the decisions we make. It can come from many different sources, but things like curiosity, personal values, and a sense of control are really important for learning and making good choices. But remember, motivation isn't only about what we feel inside. The social setting, especially when there's competition, can really boost our performance by making us pay more attention and put in more effort. Studies show that competition and comparing ourselves to others can be strong motivators, even if there aren't any clear rewards [1] [2]. Research in neuroscience has started to look at how different areas of the brain like the prefrontal cortex, insula and angular gyrus [3] respond based on whether we're motivated by our own interests or by pressures from the outside, like competition or outside rewards.

In this context, a previous study explored how implicit social pressure could influence motivation within a controlled experimental setting [4]. The researchers looked at how participants' motor decisions and brain activity changed when facing a virtual partner, believed to be either more or less skilled. Interestingly, the presence of a strong partner created a sense of social hierarchy that significantly boosted participants' performance demonstrating a clear motivational effect [Fig. 1].

The study checked out EEG recordings and saw some interesting changes in brainwave frequencies, especially in the beta (15–32 Hz) and gamma (40–80 Hz) bands. These changes showed how motivation appears in the brain through different activity patterns [5] [6] [7] in areas like the prefrontal, premotor and occipital cortices [Fig 3]. This backs up the idea that intrinsic motivation involves different neural networks [8]. Plus, machine learning methods like multinomial logistic regression (MLR) could classify motivational states with an accuracy of up to 95% in the gamma band [Fig 2], showing how useful these neural recordings can be.

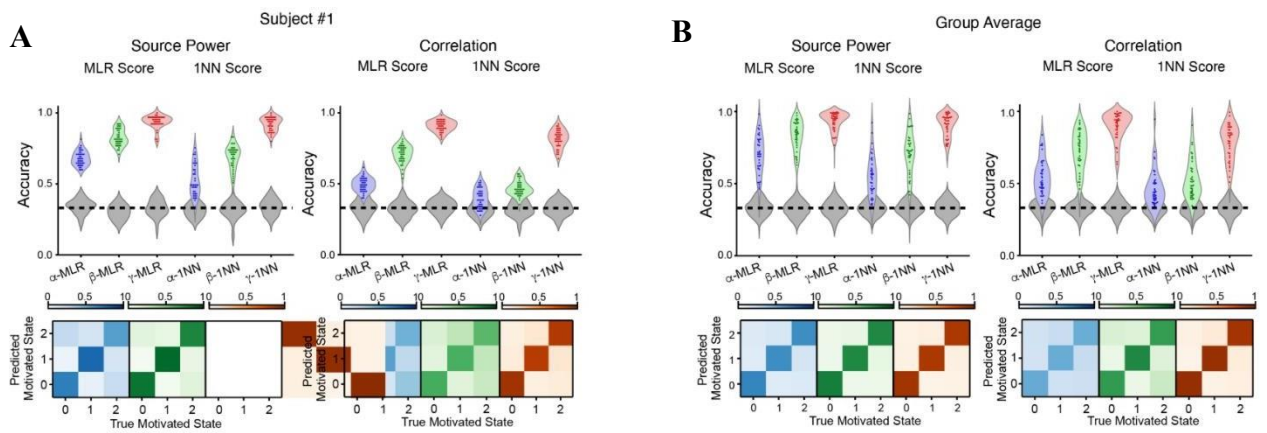
Building on this work, our study had two main goals. The first was scientific: we wanted to investigate whether a similar motivational mechanism could be observed in individuals with Parkinson's disease. Parkinson's not only causes motor dysfunction but also impacts cognitive and motivational abilities [9] [10], often due to disruptions in the dopaminergic fronto-striatal pathways.

Figure 1 :



Behavioural effect of the motivated state on the z-scored PV, PD, TTPV, DT and End-point Error. Their group average distributions have been boxplot for each motivated state (Red: Solo; Green: Easy; Blue: Hard) (*Ignasi Cos & al.*)[4].

Figure 2 :



All Sources Space EEG Analyses. A. Motivated State classification accuracy and confusion matrices, as by the MLR and 1-NN classifiers, as a function of #Motivated state, for each frequency band and metric (electrode power, correlation), performed on the EEGs of Subject #1. **B.** Same as A, averaged across all subjects. (*Ignasi Cos & al.*)[4].

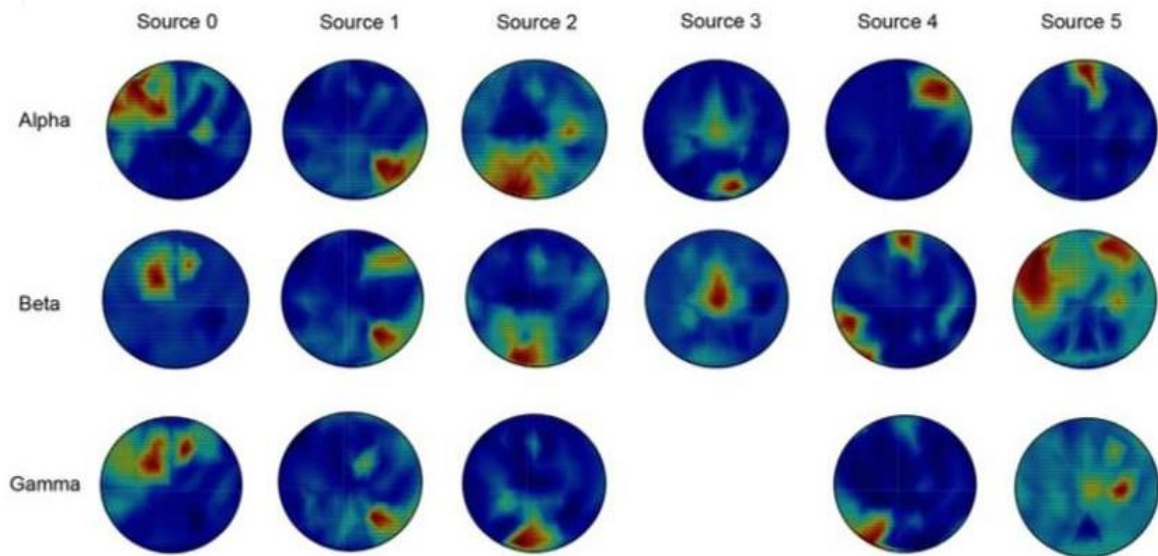
To explore this, we adapted the original experimental protocol and applied it to a Parkinson's patient. The subject was asked to complete a motor decision-making task under social pressure, with and without the presence of a virtual partner. Brain activity was recorded using high-density EEG (128 electrodes), and we focused our analysis on three key frequency bands alpha (8–12 Hz), beta (15–32 Hz) and gamma (40–80 Hz) known to be linked to attention, motor preparation, and fast sensorimotor integration [11] [12].

The second goal was more technical: we wanted to create a new EEG analysis tool adapted for Parkinson's patients. We also planned to test this updated tool on the healthy participants from the original study [4]. The old method had some drawbacks. We found that the Independent Component Analysis (ICA) with a spherical head model used before made it hard to clearly see which brain areas were involved in motivation, especially in the beta band in source 5 and the alpha band in source 0 [Fig 3].

In our pipeline, we updated our model inversion with MNE-Python and some newer tools. We added a general head model with a generic morphology. This helps us for better alignment between patients and eventually healthy controls as well. Our goal was to make the anatomical details clearer and improve accuracy in figuring out where motivation happens in the brain.

Through both the scientific and technical aspects of this study, our goal is to contribute to a deeper understanding of how motivation operates in the brain even in the context of neurodegenerative conditions and to lay the groundwork for developing EEG-based biomarkers that could one day inform therapeutic interventions.

Figure 3:



Selected Common Sources Space EEG Analyses. Common sources across subjects related to the task for the α , β and γ -bands. (*Ignasi Cos & al.*)[4].

Materials and Methods

My contribution

In this project, I was responsible for the full development of the EEG data processing pipeline. This included the entire preprocessing workflow, as well as the decoding analyses carried out both at the sensor (electrode) level and in the source space. I implemented and adapted the necessary steps to ensure that the data were clean, properly aligned with the standard head model, and suitable for classification analyses in different motivational conditions.

Participant

The data collected in this study comes from a patient diagnosed with Parkinson's disease, according to the criteria established by the MDS [13]. The participant was informed about the goals of the study and gave informed consent. The study follows current guidelines and regulations. The ethical protocol was approved by the Clinical Research Ethics Committee (CEIC-Parc Salut MAR) of Pompeu Fabra University - Hospital del Mar, reference number 2015/6085/I.

Experimental Task

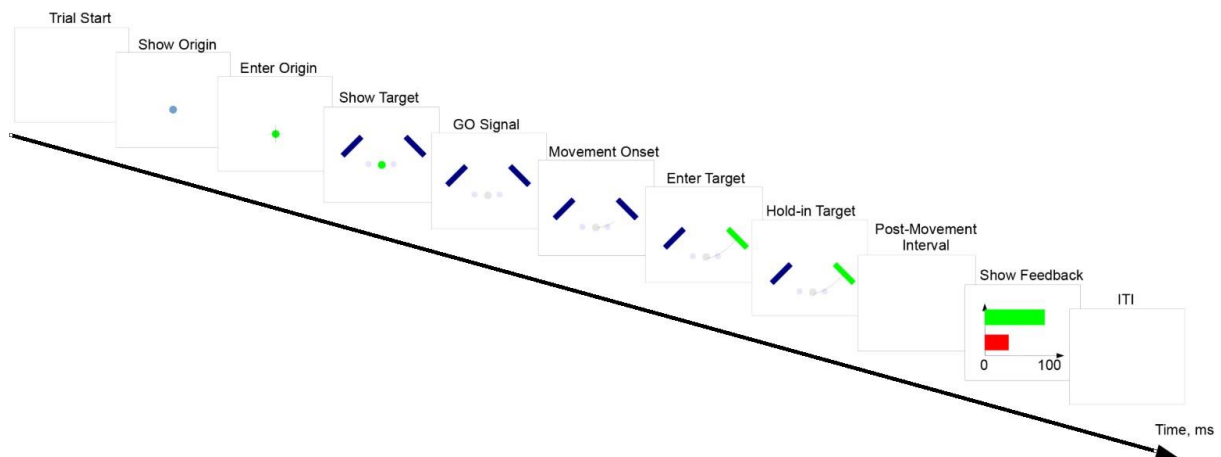
The participant performed a decision-making task involving two types of movement, the goal was to hit the target with the most precision as he can. This task was made to explore the effects of intrinsic motivation on decision-making in the context of Parkinson's disease. The movement was based on an existing protocol used with healthy subjects [4].

The task included two motor modes: "Stop-in," where the participant had to stop on the target with precision, and "Punch-Through" [Fig. 5], where the participant passed through the target and stopped freely afterward.

Each trial started with the display of a starting point, which the participant had to follow. After a delay of 800 ms, a GO signal appeared, telling the participant to start the movement to reach the target accurately [Fig. 4].

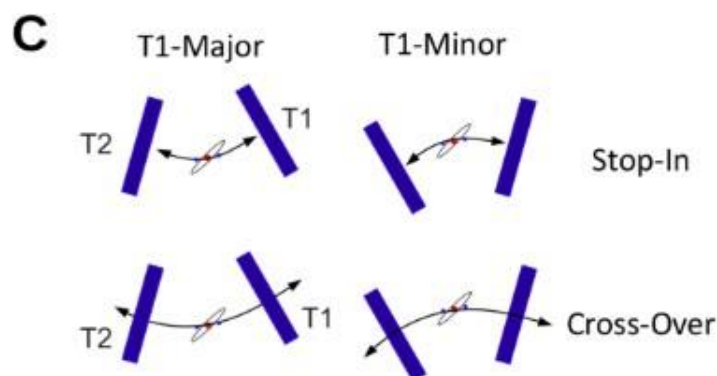
Trials were grouped into blocks, according to three motivational conditions (0–Solo, 1–Easy, 2–Hard) [Fig. 6] and the two motor modes (Punch-Through / Stop-In). In total, 12 blocks of 108 trials each were completed. The blocks were indexed as follows: even numbered blocks (2, 4, 6, 8, 10) were of the Stop-In type, and odd numbered blocks (1, 3, 5, 7, 9) were of the Punch-Through type. After assigning the blocks to movement types, we applied the motivational conditions. The blocks were labeled as: 1,1 and 2,2 for type 0–Solo; 3,4,7,8 for type 1–Easy; and 5,6,9,10 for type 2–Hard. Sessions were held on consecutive days to reduce fatigue.

Figure 4 :



Timeline of a typical trial. It starts with the presentation of a white screen during 500ms. After this time, a 1cm diameter pale blue origin circle cue is presented on the centre bottom part of the screen. 1ms after the subject's fingertip crossed its border, the origin cue changed to green, and two rectangular blue targets (10 x 1cm) and two pale blue via-points (1cm diameter) were presented to the right/left of the origin cue. Path distances from the origin to either target extended 15cm. The GO signal was reported by removing the origin cue, at which time the participant could report his/her choice by moving the cursor from the origin to the selected target. The rectangle target turned from blue to green when the cursor entered the target. After 500ms, everything disappeared. 500ms after this, a green horizontal bar was shown during 500ms to report the precision of the movement just performed. If the participant was accompanied, a red bar was also shown below, reporting the simulated partner's precision. An empty screen ITI followed, the duration of which extended the trial to its 7000ms fixed duration. (Ignasi Cos & al.)[4]

Figure 5 :



Geometrical arrangements (T1-Major/T1-Minor) and control configurations (STOP-IN/CROSS-OVER). (Ignasi Cos & al.)[4]

Motivation manipulation

Intrinsic motivation was manipulated by introducing an implicit social hierarchy. The participant was informed that, in each block, a “partner” from a community of other participants would be present. In reality, the partner was virtual. Their performance level was controlled to be either higher (Hard condition) or lower (Easy condition) than that of the participant.

When a virtual partner was present, a red bar was also displayed, showing the partner’s performance for the same trial. Then, every nine trials, after the precision bars [Fig. 4], the participant’s ranking compared to the partner was shown. Additionally, a photograph of the participant and their partner was presented, ranked from top to bottom, with a height proportional to their average final precision.

Even though it was clearly stated that this was not a competition, the introduction of this social bias was intended to implicitly modulate the participant’s motivation by increasing their attention and motor engagement. We also matched the participant’s and the partner’s gender to avoid gender-related effects.

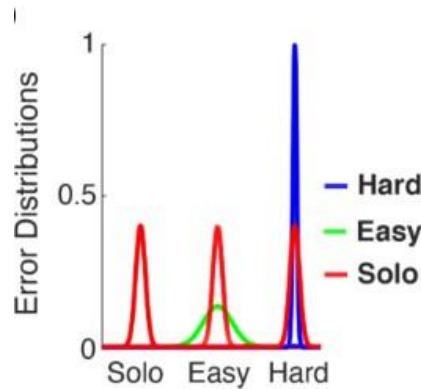
EEG Preprocessing

We chose a 1600 ms time window for further analysis, starting 800 ms before the first stimulus and ending 800 ms after. Why this interval? It allows us to analyze premotor signals and to look for early changes in the motivation network. Moreover, since this window precedes any movement, it likely contains fewer artifacts than those occurring after the stimulus.

EEG signals were recorded using a 129 electrodes system (GSN129) with a sampling rate of 500 Hz. Signal preprocessing (at both the electrode and source levels) followed a defined pipeline [Fig. 7]. We loaded the data, which had already undergone some preprocessing steps, such as a notch filter around 60 Hz was also applied to eliminate electrical interference from the power supply.

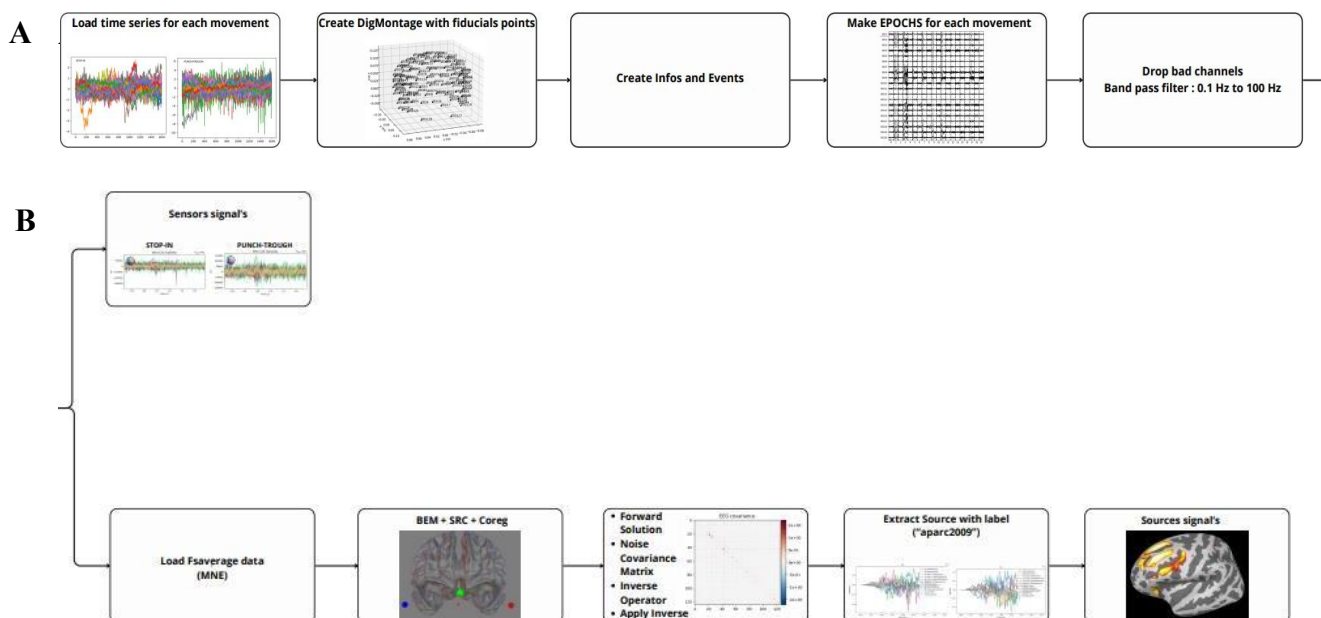
We obtained two time series instances, one for each movement (SI and PT) [Fig. 8]. Trials with NaN values were excluded. As a result, for each movement, we had a four dimensional instance like this: (128, 1600, 107, 6); (n_channels, n_times, n_trials, n_blocks).

Figure 6 :



Depiction of the distributions of error associated to our artificial partners (Green: Easy - broad variance; Blue: Hard - narrow variance), and of a participant (Red)(*Ignasi Cos & al.*)[4]

Figure 7 :



A. General Processing Pipeline : The pipeline begins with loading time series data for various movements, followed by creating a digital montage with fiducial points. Events and relevant information are then defined, and the data is segmented into epochs for each movement. The final step involves removing bad channels and applying a band-pass filter (0.1 Hz to 100 Hz).

B. Getting Sensor and Source Signals: Sensor signals that show different movements and load the Fsaverage data for source localisation. We set up a Boundary Element Model (BEM) and do source reconstruction and alignment (coregistration). Then, the forward solution, noise covariance matrix, and inverse operator were calculate to pull out and label source signals using a specific scheme. The final output shows the source signals and gives us a look into brain activity for each movements (Stop-In and Punch-Trough).

Then, we applied the montage to assign a spatial configuration to the electrodes, using the GSN129 system [Fig. 9]. We also added the fiducial points (nasion, left and right preauricular points), which are necessary for later steps. Following the predefined pipeline using MNE-Python, we created the “info” instance that includes metadata (channel names, sampling frequency, signal types, etc.).

In addition, we created “events” to mark the conditions of stimuli or specific actions during the recording. These were not present in the .mat file, so we created them ourselves. To do this, we made an array for each motivational condition with 214 trials per condition, and a time step of 1600 ms, since our time resolution is 2 ms.

Next, we segmented (epoched) the data based on the motivational events (solo, easy, hard) [Fig. 10]. Each trial was isolated in a time window centered on the event, which allows accurate time and frequency analyses for each condition.

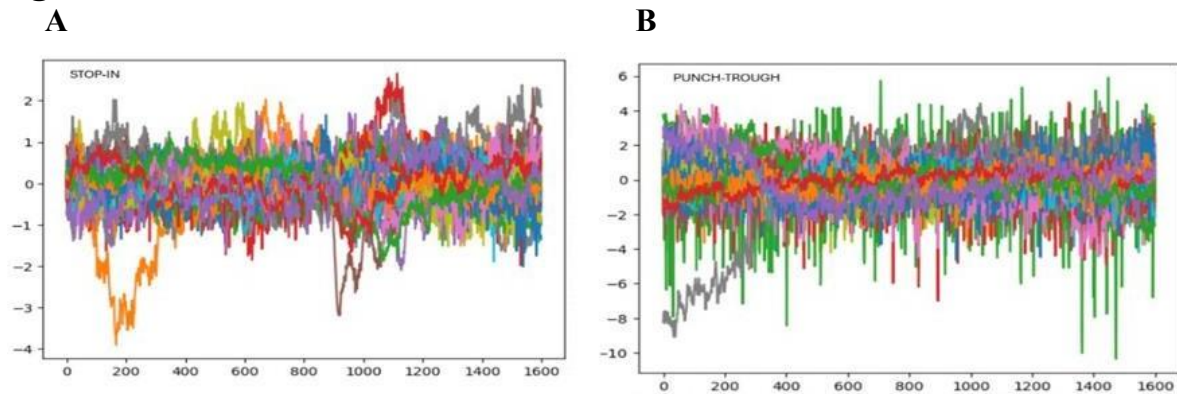
After getting the epochs for each condition and each movement, we applied a filter to limit the frequency range between 1 Hz and 100 Hz. Corrupted channels (EEG94, EEG99, EEG127) were also removed. This step involved detecting and rejecting electrodes with too much noise or signal loss, based on visual and statistical criteria (like abnormal variance or saturation).

Finally, for each movement, we merged the epochs of the three conditions (3 epochs per movement) into one large epoch. This was done to avoid losing signal quality or adding noise. If we had used one large epoch per movement without the original events (since we created them manually), the analysis would not have been reliable. That is why we chose to filter small condition-specific epochs first, and then merge them at the end to get one large epoch for each movement: Stop-in and Punch-Trough.

Extraction of Neural Signals (Source Space)

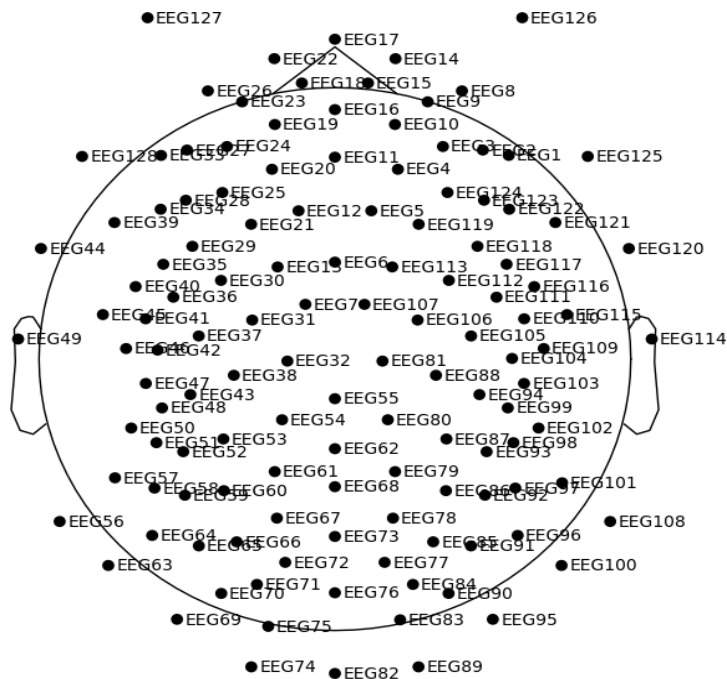
To extract source signals from the EEG data, we followed a processing pipeline using the MNE-Python toolbox [Fig. 7B]. First, we loaded the standard brain anatomy “fsaverage”, which provides a common morphological reference for all analyses. Then, we performed the BEM (Boundary Element Method) modeling, defined the cortical surface (“SRC”), and carried out the coregistration between EEG data and anatomy [Fig. 11]. These steps are needed to build the forward model. Once done, we calculated the forward solution and the noise covariance matrix (“noise cov”).

Figure 8 :



Times series for each movement : **A. STOP-IN:** This panel displays the time series data for the "STOP-IN" movement. The plot shows multiple colored lines representing different channels, capturing the variations in signal amplitude over time. **B. PUNCH-THROUGH:** This panel presents the time series data for the "PUNCH-THROUGH" movement. Similar to panel A.

Figure 9 :



The spatial configuration of the GSN129 electrode setup as defined by the MNE Python library. Each labeled point corresponds to the position of an individual EEG electrode on the scalp, providing a comprehensive layout of the electrode arrangement used in the GSN129 system.

These allowed us to create an inverse operator, which estimates cortical activity from the EEG signals recorded on the scalp. Using this inverse operator, we extracted the source signals based on anatomical labels from the "aparc2009" atlas. This extraction helped isolate activity from specific regions of interest, giving us a time series of source signals that we later visualized and analyzed [Fig. 12].

EEG Decoding in Electrode and Source Spaces

We took two different approaches: one looked at the electrode level and the other at the source level [Fig. 13]. We filtered the raw EEG signals into three frequency bands: Alpha (8-12 Hz), Beta (15-32 Hz), and Gamma (40-80 Hz) using bandpass filters. This helps us focus on brain activity related to different cognitive tasks

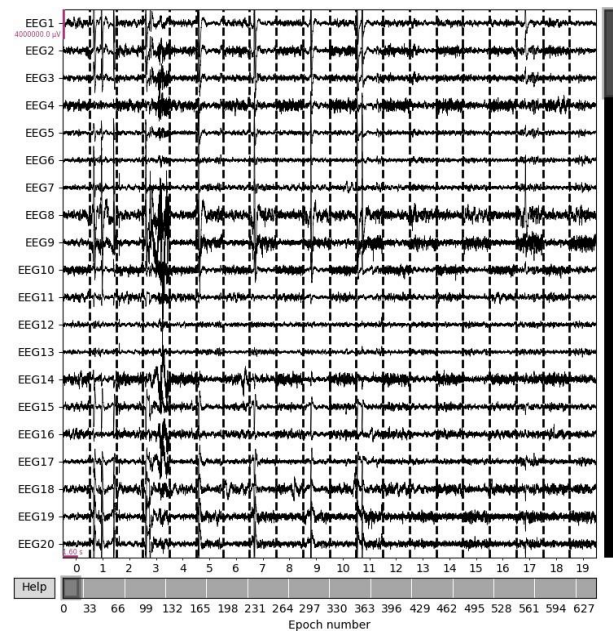
In the electrode space, we computed the power spectral density (PSD) [14], for each electrode and frequency band using the “multitaper” method, focusing on a time window from -0.8 to 0.0 seconds to analyze the pre-movement period. The multitaper method uses multiple orthogonal windows called “tapers” (often based on Slepian functions) to estimate the spectrum from the same signal segment. The results are averaged to reduce variance while keeping good frequency resolution.

The PSD helps us see how the power of a signal is distributed across different frequencies. In EEG analysis, this is useful for spotting the main brain frequencies. We used it to pull out features from the EEG signal. With that, we could start decoding the signals, looking at each electrode individually using some classification methods that we’ll explain later. But we wanted to take it a step further and check out the interactions between pairs of electrodes.

To do this, we first applied the Hilbert transform [15] [Fig. 14] for each frequency band to get a better signal analysis. This transformation gives us a real part (the filtered original signal) and an imaginary part. This allows us to extract the signal envelope, which helps estimate the instantaneous oscillatory activity and phase, useful for studying signal synchronization. This is especially helpful in EEG studies of functional connectivity or responses to stimuli.

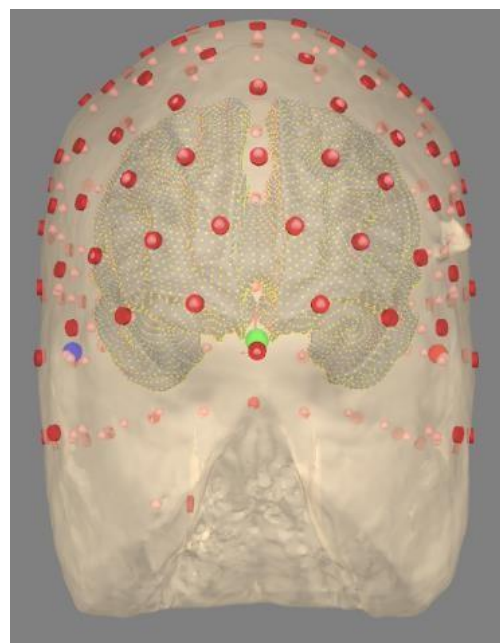
We then used these envelopes to calculate correlation matrices between pairs of electrodes [Fig. 15A], representing the functional links between brain regions. For each frequency band and each movement, we vectorized these correlation matrices by selecting only the semi-diagonal elements, which were used as feature vectors for input into our classification models.

Figure 10 :



Epochs Data for Movements and Conditions. This figure presents the EEG epochs data segmented for different movements, specifically "STOP-IN" and "PUNCH-THROUGH," under various conditions labeled as solo, easy, and hard. Each row represents the signal from a different EEG electrode, with time on the x-axis and signal amplitude on the y-axis.

Figure 11 :



Coregistration of EEG Channels. This figure illustrates the coregistration process between EEG channels and a standard model of the head scalp. The red markers represent the positions of the EEG electrodes, which are aligned with the anatomical landmarks of the scalp model.

In the source space, the signals were reconstructed using the inverse solution [16], based on the *aparc2009* atlas, giving time series for each cortical region of interest. As with the electrodes, correlation matrices [Fig. 15B] were calculated between these regional time series. These matrices give a picture of functional connections at the cortical source level. To better understand frequency-specific relationships, we also performed a spectral coherence analysis [17]. This method measures how synchronized two regions are in terms of frequency, which is more precise than simple time correlation.

The goal of our classification analysis was to predict the experimental condition using the features extracted from EEG signals, both in electrode and source space. We used two types of information to describe the brain's functional network: oscillatory power (from PSD) and inter-regional relationships (correlation or coherence between signals, as indicators of neuronal interactions). All trials ($n = 642$ per movement) were randomly split into 80% for training and 20% for testing, and this procedure was repeated 100 times to robustly and reliably estimate the classifiers' ability to generalize [18].

For the correlation analysis, we took the average from all the trials. We divided the data into blocks of 20 trials and did this 10 times, resulting in 22 average segments for each condition. That gave us a total of 66 segments for the movement task (Stop-in or Punch-Through). This approach let us apply the inverse solution to each average segment without overloading our computations.

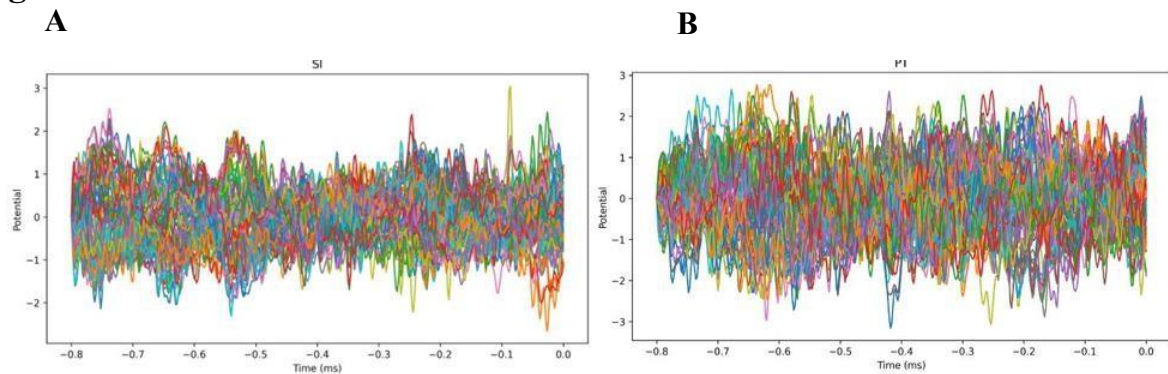
We wrote three algorithms with some custom Python scripts, using open-source libraries like *numpy*, *scipy*, and *scikit-learn* [19] :

Multinomial Logistic Regression (MLR) [20], a linear classifier that learns a weighted combination of features to best separate the classes. The model learns by adjusting the weights to minimize prediction errors. The output is probabilistic, allowing us to estimate the model's confidence for each class.

k-Nearest Neighbors (k-NN) [21] is a simple method where we assign a label to a new sample based on the label of its closest match in the training data. Since it doesn't need a training phase, it's easy to use, but it can be pretty sensitive to how the data is arranged nearby.

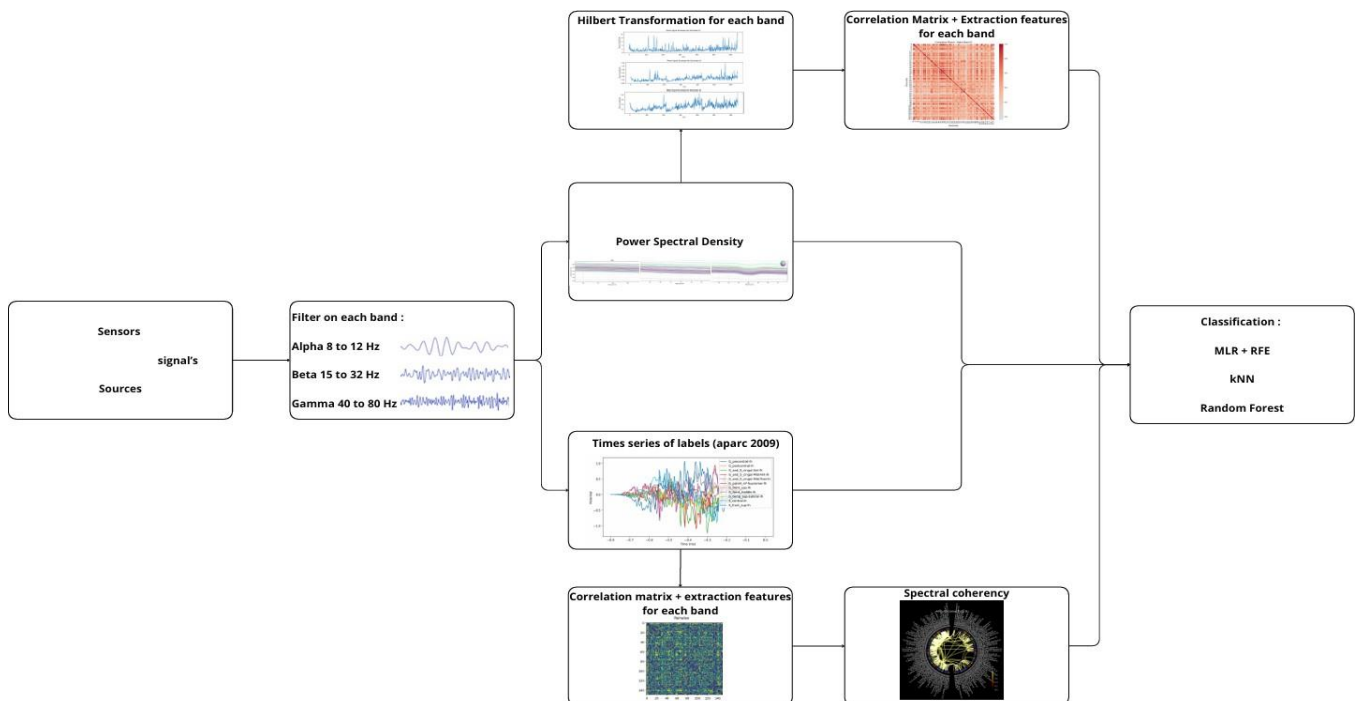
Random Forest [22] is a technique that combines multiple decision trees to make predictions. Each tree is trained on a random subset of the data, and the final decision comes from the majority vote of all the trees. It's pretty effective with messy data and can identify which features matter the most. To find the most discriminant features, we applied Recursive Feature Elimination (RFE) [23] using the MLR coefficients. At each step, the least important features were removed until we reached an optimal subset.

Figure 12 :



The extracted time series data of Regions of Interest (ROIs): **A. STOP-IN:** This panel shows the time series filtered from [Fig. 8A] data for the "STOP-IN" movement. The plot displays the potential variations over time for different ROIs, with each line representing a different source or trial. **B. PUNCH-THROUGH:** This panel presents the time series filtered from [Fig.8B] data for the "PUNCH-THROUGH" movement. Similar to panel A.

Figure 13 :



Pipeline for Decoding Power or Correlation in Sensors/Sources Space. The pipeline used for decoding power or correlation in the sensors and sources space of EEG data.

Results

EEG decoding in sensor space based on PSD

The spectral power of the electrodes allowed us to test different decoding models [Fig 16], highlighting conditions that change depending on the movement type and the frequency band. For the Stop-in movement, the performance obtained with different classifiers showed a good capacity to predict motivational states.

We used Multinomial Logistic Regression (MLR) and noticed that accuracy went up as we moved from the alpha band to the gamma band. We saw 82% accuracy in alpha, 87% in beta, and 90% in gamma. This points to the idea that faster brain waves, especially in the gamma range, might give us some good clues about motivation before people take action. When we checked with Random Forest, the results were pretty similar but a little different. It had 80% accuracy in alpha, 87% in beta, and 88% in gamma, which supports the thought that higher frequencies are more informative. On the flip side, the k-Nearest Neighbors (kNN) classifier didn't do as well overall, with just 71% accuracy in alpha, but it did improve to 82% in beta and 81% in gamma.

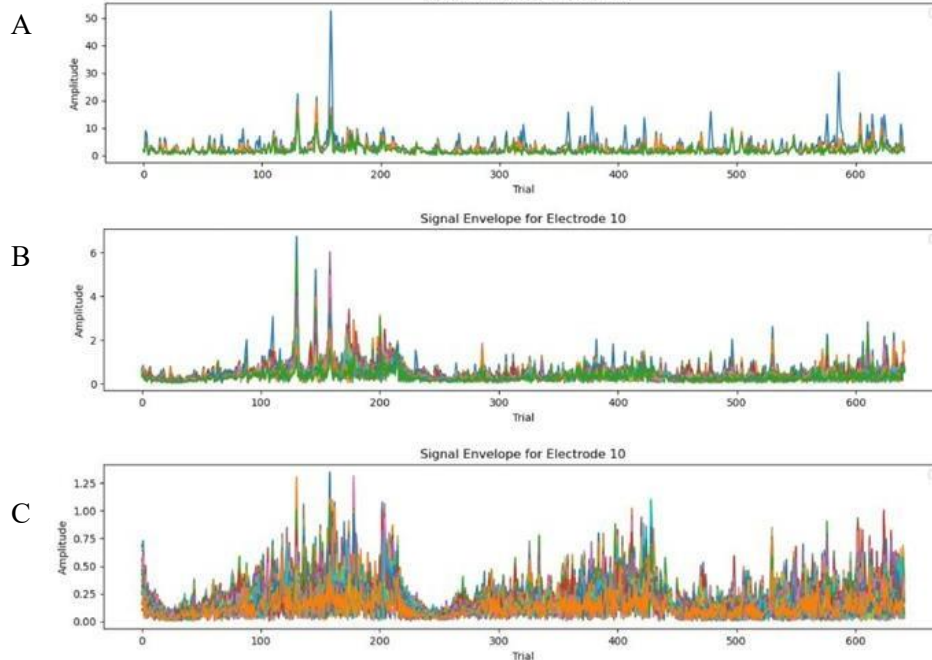
For the Punch-Through movement, all classifiers showed higher accuracy scores across all frequency bands. MLR remained stable with 82% in alpha, improved to 88% in beta, and stayed at 90% in gamma, which is similar to the Stop-in movement.

Random Forest performed best, reaching 80% in alpha, 90% in beta, and 93% in gamma. This increase in performance may indicate more specific brain activation for a movement that is perhaps more free or natural. Finally, while kNN remained less accurate than the others, it still improved notably, especially in beta (85%) and gamma (84%).

It seems that the type of classifier you pick might not matter that much since all the models showed similar results across different frequency bands

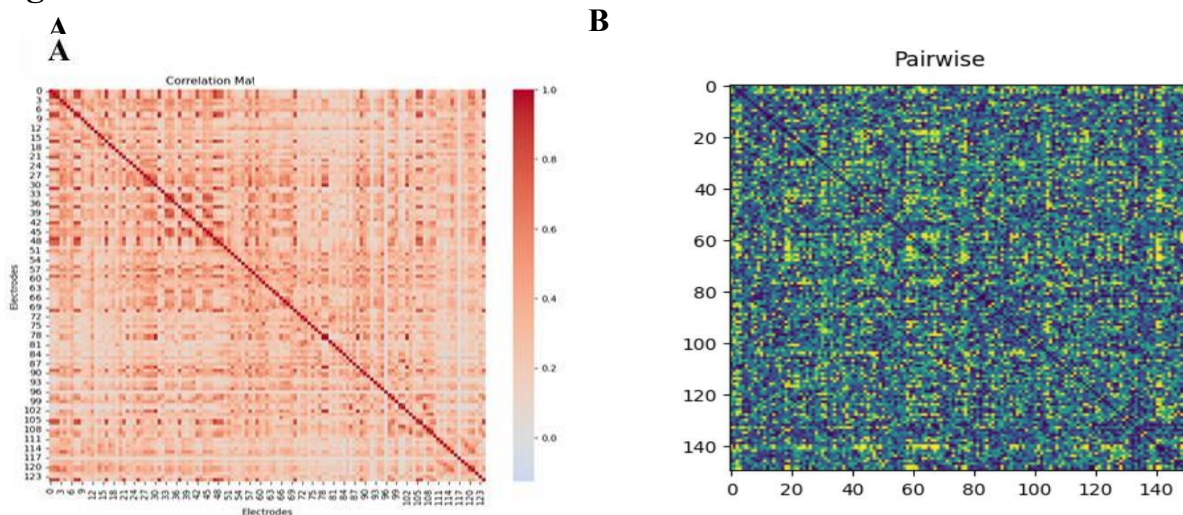
The Recursive Feature Elimination (RFE) method [Fig 17], applied to the gamma band and the MLR classifier, helped us find the ten most informative electrodes for each movement. In the Stop-In movement, the key electrodes were mostly found in the central and back areas of the brain, which hints at the involvement of motor and parietal regions. For the Punch-Through movement, there were some differences, but a few electrodes were active in both movements. This could suggest that there are shared motivational factors when preparing for movement in Parkinson's patients.

Figure 14 :



Hilbert Transformation Envelopes for EEG10 Channel: A. Alpha Band (8 to 12 Hz) B. Beta Band (15 to 32 Hz) C. Gamma Band (40 to 80 Hz): Each plot shows how the neural oscillations change in strength for the EEG10 channel across different trials, giving a look at the timing patterns

Figure 15 :



Correlation matrices for EEG data: A. Sensor's Space: Displays the correlation matrix among various electrodes in the sensor space, illustrating the relationships and interactions between different EEG channels. **B. Source's Space:** Shows the pairwise correlation matrix in the source space, providing insights into the connectivity and interactions among different brain regions.

Electrodes pairs decoding

To go further in our spatial and functional analysis, we tested electrode pairs using power spectral density (PSD) features for each frequency band. This helped us see if some combinations of electrodes were more helpful in predicting motivational states, depending on the frequency band and the movement.

We explored the functional structure behind premotor brain activity by measuring the synchronization between scalp regions. Using the same implementation protocol (see Materials and Methods), we created correlation matrices, kept only the upper triangle, and used it as a feature vector for supervised models, tested for each movement. This showed different patterns depending on the frequency.

With the MLR model [Fig 19], we used the usual 80/20 split for training and testing. We found that accuracy improved with different frequency bands. The alpha band had 44% accuracy, beta went up to 50%, and the gamma band performed the best at 56%. This indicates that quick syncing in the brain is key for getting ready to move, especially in the Stop-in condition. We noticed a similar pattern in the Punch-Through movement, where all frequency bands had even better performance.

In the alpha band, correlations stayed low, with accuracy around 42%. But performance improved in the beta band around 54% and reached its best in gamma, with 62% accuracy. This pattern, similar to the results from spectral power features, supports the idea that high-frequency activity carries more discriminative information about the motivational state during the more free movement (Punch-Through).

For the kNN model [Fig 19], the performance trends were similar to those seen with PSD features, for both movements and all three frequency bands. The scores were lower and less discriminative, but again, the Punch-Through movement showed a small improvement across bands. The beta band gave the best result with 45% accuracy. However, for the Stop-In movement, this time the gamma band was not the most efficient. The three bands had similar accuracy scores, all around 45%.

Figure 16 :



The Power Spectral Density (PSD) and classification performance for two movements, “STOP-IN” and “PUNCH-THROUGH,” across Alpha, Beta, and Gamma frequency bands using three classifiers: Multiple Logistic Regression (MLR), k-Nearest Neighbors (kNN), and Random Forest (RF). The violin plots at the top of each section depict the PSD distribution for each frequency band and movement type. Below these plots, confusion matrices as heatmaps show the classification results, providing insights into the accuracy and performance of each classifier in distinguishing between the movements.

For the Random Forest model [Fig 19], the results were close to those of MLR, with a small difference in the Punch-Through movement, where the beta band had better performance (>50%) than gamma. The alpha band still gave the lowest results for both movements, confirming that high-frequency bands seem to play a major role in decoding motivational states. For the Stop-In movement, the ranking of bands matched earlier results, with gamma highest (53%) and alpha lowest (45%).

These results suggest that looking at how electrodes interact, particularly in the gamma band, can give us a better grasp on motivation. It seems that analyzing correlation data provides more insights than just focusing on the signal strength by itself.

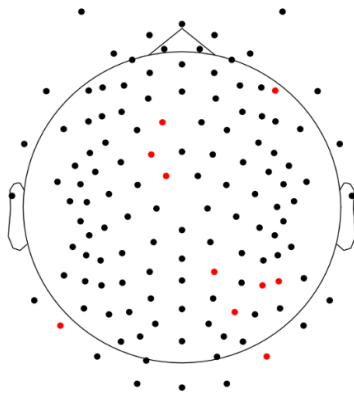
To support the quantitative findings, we also visualized the ten most informative functional connections [Fig 18] using the MNE-Python library. These were selected from the gamma band, separately for each movement type (Stop-In and Punch-Through), helping to show the neural dynamics behind these two motor conditions.

For the Stop-In movement [Fig 18A], the most discriminative functional links were spread bilaterally, involving connections between temporal, parietal, and frontal electrodes, especially between EEG68, EEG17, EEG65, and EEG69. This pattern might reflect the integration of motor and attentional control networks, which are needed to stop the movement accurately on the target.

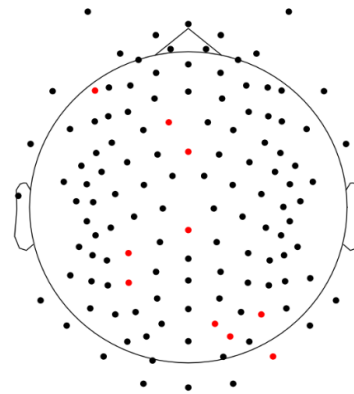
In contrast, for the Punch-Through condition [Fig 18B], although we found similar connections (especially repeated links like EEG17–EEG68 and EEG21–EEG68), there was more dense activity in the right parieto-occipital and temporal regions. The recurring presence of certain pairs (like EEG65–EEG45 or EEG69–EEG63) in both movements suggests strong functional areas that are consistent across motor conditions.

Figure 17 :

A



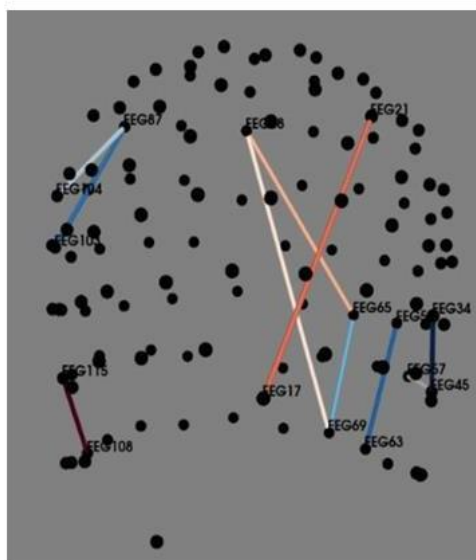
B



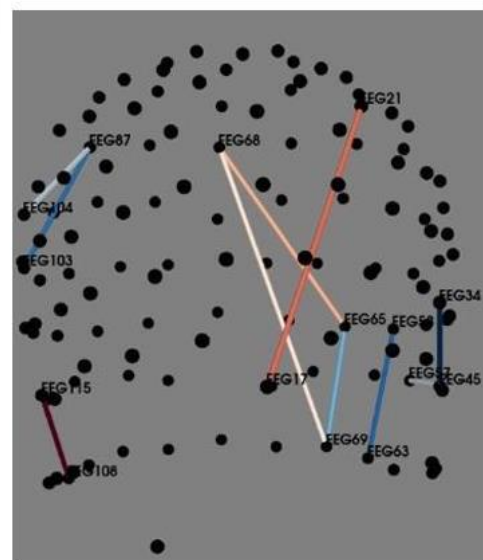
The spatial distribution of the most significant electrodes for the “STOP-IN” and “PUNCH-THROUGH” movements, with the top-performing electrodes highlighted in red. A “STOP-IN” movement, with the best features indicated by the indices [85, 97, 79, 12, 95, 63, 7, 92, 91 et 112]. B “PUNCH-THROUGH” movement, with the best features indicated by the indices [60, 84, 6, 53, 12, 33, 95, 91, 55 et 40].

Figure 18 :

A

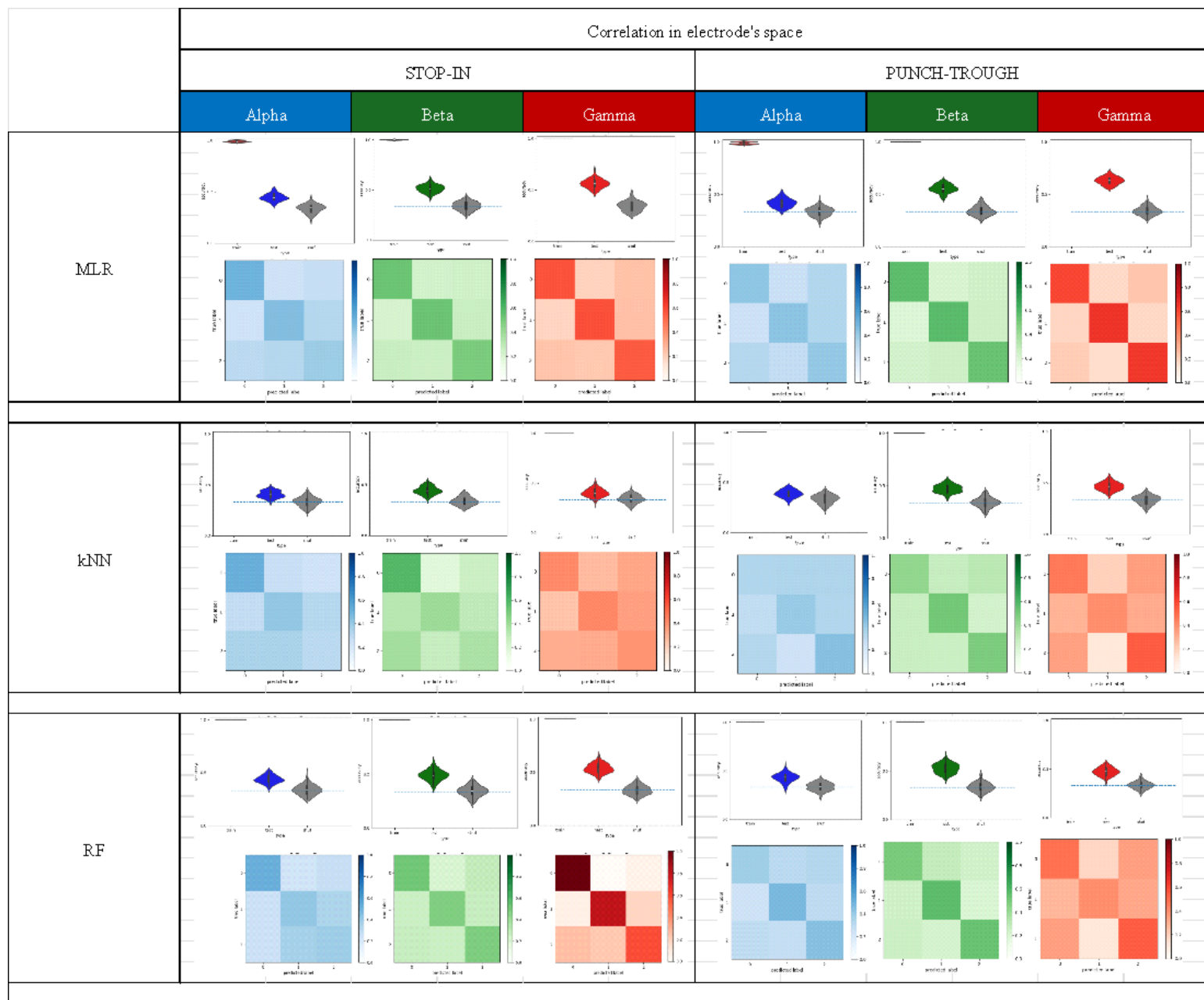


B



Functional connectivity patterns between the most informative electrode pairs in the gamma band, identified using MLR classifier weights. A. Stop-in movement condition; B. Punch-Through movement condition. Colored lines indicate the strength and direction of the discriminative connections, with warmer colors representing higher contribution to classification.

Figure 19 :



The Correlation per pairs of sensors and classification performance for two movements, “STOP-IN” and “PUNCH-THROUGH,” across Alpha, Beta and Gamma frequency bands using three classifiers: Multiple Logistic Regression (MLR), k-Nearest Neighbors (kNN), and Random Forest (RF). The violin plots at the top of each section depict the correlation distribution for each frequency band and movement type. Below these plots, confusion matrices as heatmaps show the classification results, providing insights into the accuracy and performance of each classifier in distinguishing between the movements.

Time Series Extraction and Source-Space Decoding of ROIs

As a next step in our study, we analyzed the decoding of motivational states in the source space. Using the inverse operator on the epoched data (based on anatomical labels from the *aparc2009* atlas), we extracted regional time series, which represent the neural activity in different regions of interest (ROIs). We then applied the MLR classifier.

The performance of this model was reported for each frequency band (alpha, beta, gamma) and both motor regimes (Stop-In and Punch-Through) [Fig 20].

For the Stop-In movement, the more often it happened, the better the accuracy got. The alpha band started at about 50%, then moved up to 60% in beta, and hit 85% in gamma. This means that quick syncing between brain areas in the gamma band is really important for handling motivation when preparing for tough motor tasks.

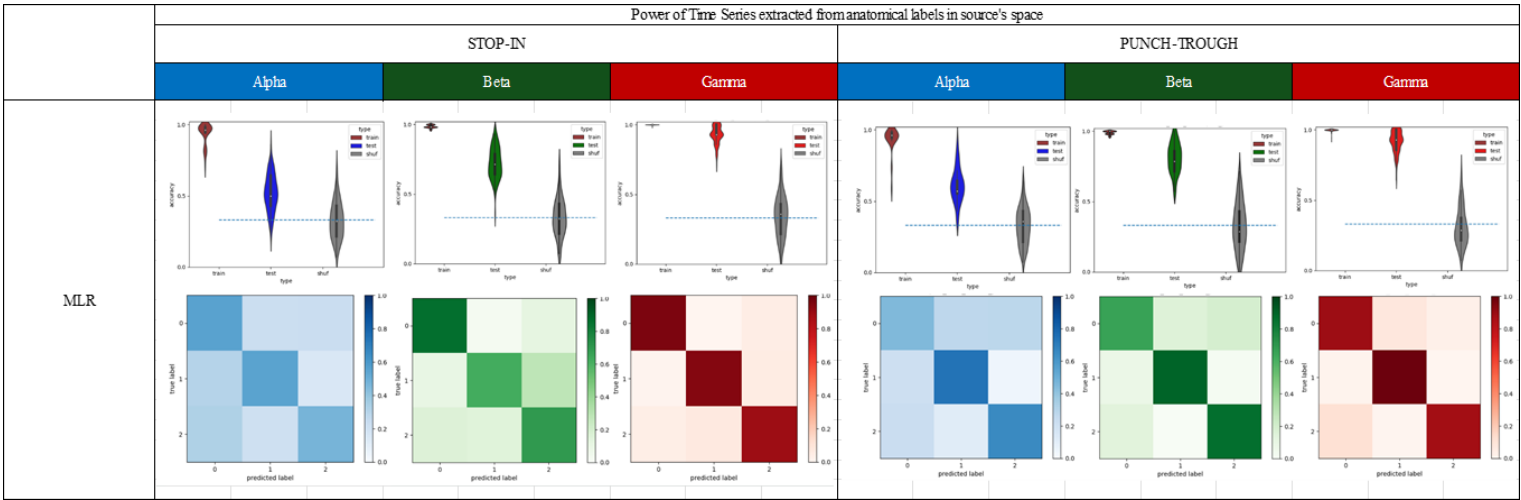
In the Punch-Through condition, the overall performance was better across all bands, consistent with the results in electrode space. Accuracy reached 54% in alpha, 73% in beta, and nearly 90% in gamma, the highest score in this part of the study. This confirms that less constrained movements (like Punch-Through) are more clearly captured by fast neural activity in cortical sources.

This is also supported by the confusion matrices, which show that the extreme motivational states (solo and hard) were better distinguished, especially in the beta and gamma bands.

To complete our analysis in the source space, and since the gamma band showed the best classification performance with MLR, we wanted to identify the most informative brain regions for predicting motivational states (solo, easy, hard). To do this, we applied a Recursive Feature Elimination (RFE) on the weights of the MLR classifier, trained only with gamma band data. This allowed us to automatically select the ten most important regions for each movement type. We then showed them on different brain views (front, back, left, and right), using colors to highlight the selected areas [Fig 21].

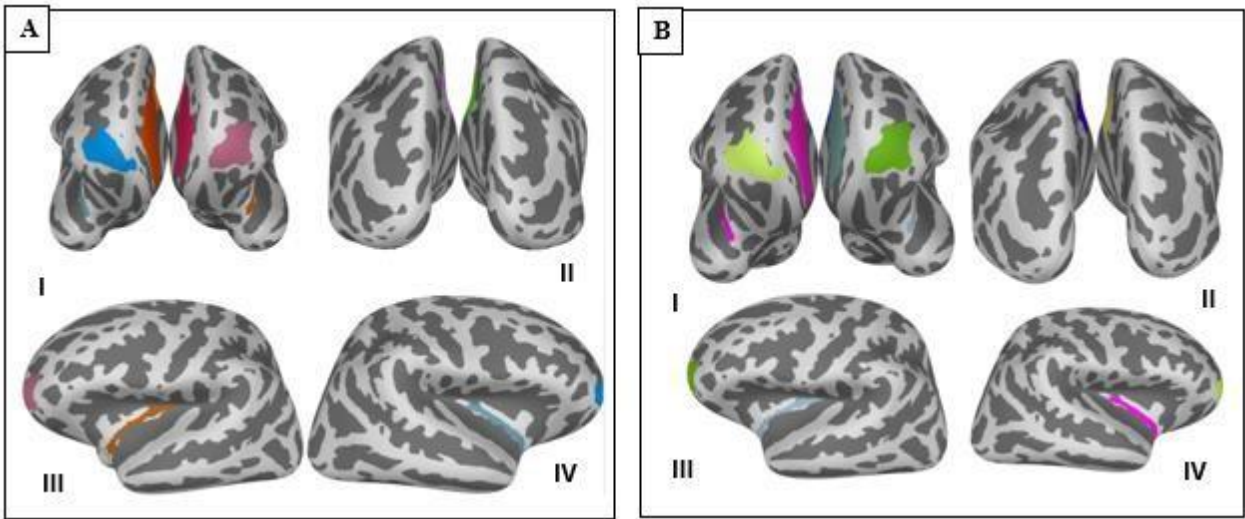
For the Stop-In movement [Fig. 21A], the most informative regions were mainly located in the cingulate cortex (anterior and mid-posterior parts), the frontomarginal gyrus, and the long insular gyrus, across both hemispheres. These areas span frontal, insular, and midline cortical regions, suggesting that during controlled movements like Stop-In, the modulation of motivational state involves a distributed network that includes motor and premotor areas (cingulate cortex), high-level associative regions (frontomarginal gyrus).

Figure 20 :



Classification performance based on the power of source-reconstructed time series across frequency bands (Alpha, Beta, Gamma), using the Multinomial Logistic Regression (MLR) classifier. Results are shown separately for Stop-in (left) and Punch-Through (right) movement conditions. Upper panels display violin plots of test accuracies across cross-validation splits, while lower panels show the confusion matrices.

Figure 21 :



Cortical representation of the top 10 regions of interest (ROIs) selected using recursive feature elimination (RFE) on gamma-band EEG source data with a multinomial logistic regression (MLR) model. Panel A shows the "stop-in" movement, and Panel B shows the "punch-through" movement. ROIs are color-coded, and four cortical views are presented for each condition: I – anterior, II – posterior, III – left lateral, and IV – right lateral.

For the Punch-Through movement [Fig. 21B], the ten most informative regions showed a strong presence of cingulate areas (anterior, mid-anterior, and mid-posterior) as well as the insular and frontomarginal regions in both hemispheres. This pattern points to a greater involvement of midline and associative areas, suggesting that in more spontaneous or freer movements like Punch-Through, motivation is more closely tied to internal decision-making, cognitive control, and emotional or interoceptive processing. These findings support the idea that this type of movement recruits a more cognitive and integrative neural network.

In summary, this source-space analysis with RFE shows that the spatial brain pattern of informative regions changes depending on the movement type. This confirms that motivation is represented in a distributed way in the brain, and that the gamma band carries rich information to capture its dynamics.

Decoding connectivity/coherence between regions of interest (ROIs)

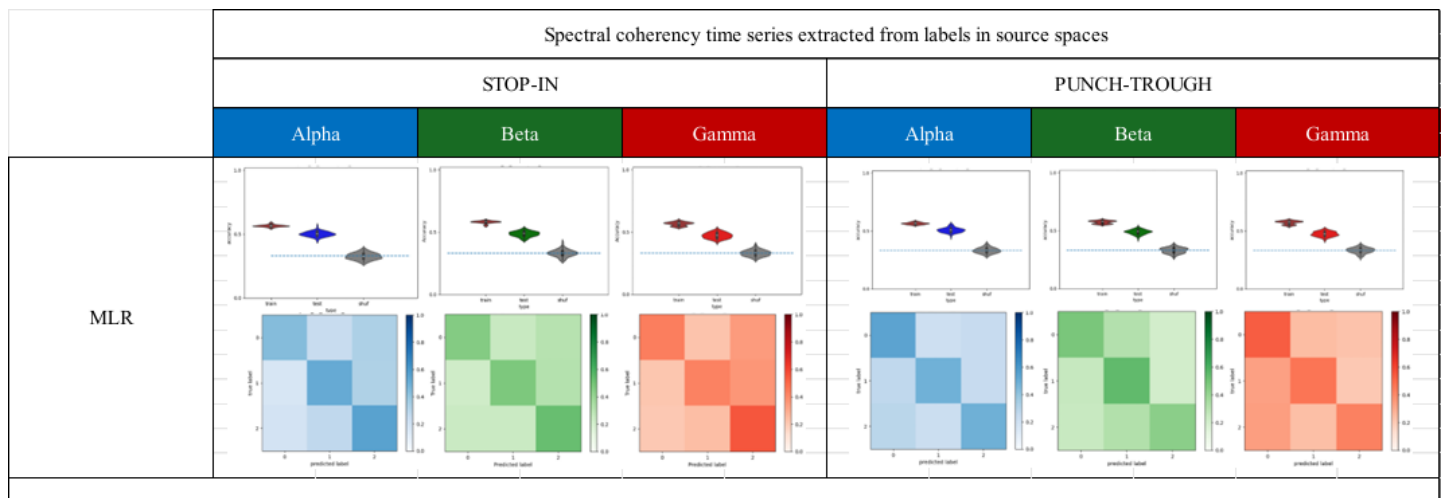
Continuing our study in source space, we further analyzed the functional interactions between cortical regions using a spectral coherence analysis. This method measures the synchronization between pairs of regions of interest, providing a more specific measure than simple temporal correlation. Coherence was calculated for each frequency band (alpha, beta, gamma) separately for the two movements (Stop-In and Punch-Through). Then, we applied the MLR classifier on the vectorized inter-regional coherence matrices [Fig 22].

The model's performance [Fig 22] confirms previous trends: a progressive increase of accuracy with frequency band, with a clear rise in the gamma band. For the Stop-In movement, average accuracies reach about 44% in alpha, 53% in beta, and go up to 60% in gamma. This increase supports the idea that fast oscillations (gamma band 40-80 Hz) play an important role in structuring neuronal activity involved in constrained motor preparation.

For the Punch-Through movement, the same pattern of increasing performance across bands is observed, with scores of 50% in alpha, 63% in beta, and peaking at 68% in gamma. This confirms earlier results: freer movements are easier to decode. Analysis of the confusion matrices [Fig 22] also shows that extreme motivational states (solo and hard) are better distinguished, especially in beta and gamma bands.

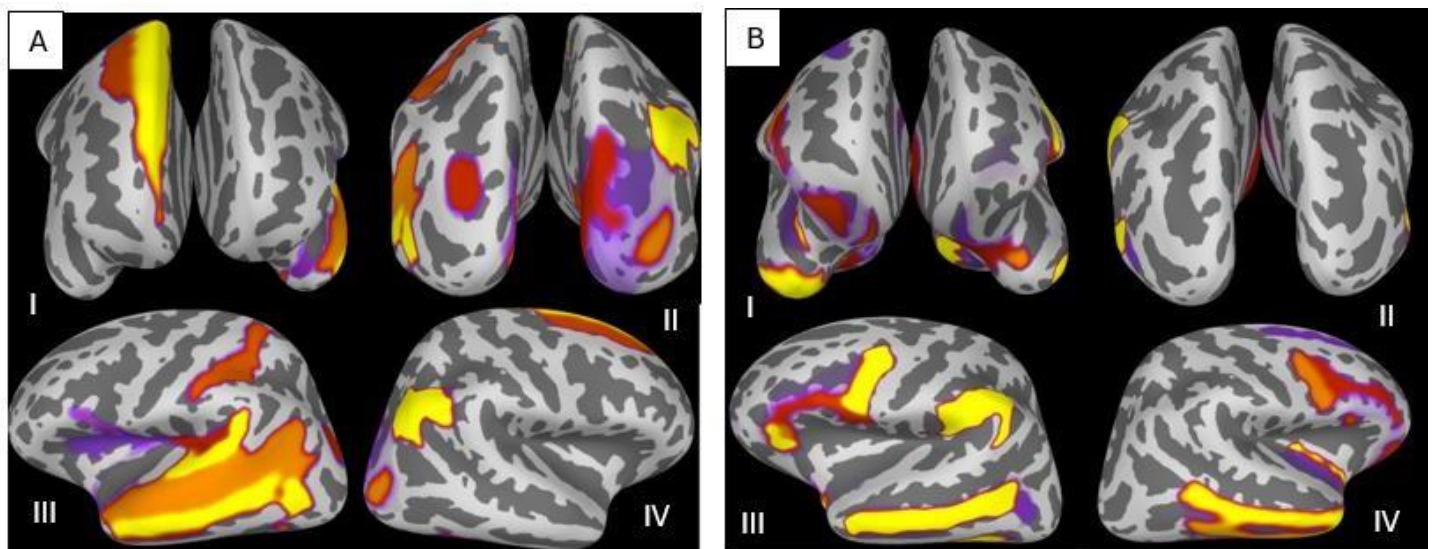
Continuing our coherence analysis, we explored the spatial topography of gamma connectivity by visualizing cortical maps of spectral coherence between regions [Fig 23]. These connectivity maps were projected on a standard brain model for each motor condition.

Figure 22 :



Classification performance based on the spectral coherency of source-reconstructed time series across frequency bands (Alpha, Beta, Gamma), using the Multinomial Logistic Regression (MLR) classifier. Results are shown separately for Stop-in (left) and Punch-Through (right) movement conditions. Upper panels display violin plots of test accuracies across cross-validation splits, while lower panels show the confusion matrices.

Figure 23 :



Cortical maps of regions showing high gamma-band connectivity, based on spectral coherence computed from EEG source time series across all labeled cortical regions. Panel A corresponds to the "stop-in" condition, and Panel B to the "punch-through" condition. Colors indicate the strength of coherence, and four cortical views are displayed for each condition: I – anterior, II – posterior, III – left lateral, and IV – right lateral.

In the Stop-In condition, we mainly see brain activity in areas like the superior frontal, mid-cingulate, posterior parietal, and occipital regions, with the left side being more active [Fig. 23A]. On the other hand, the Punch-Through condition shows brain activity that spreads out more to both sides, particularly in the anterior and mid-cingulate cortex, temporal lobe, prefrontal areas, and premotor-insular regions [Fig. 23B]. This wider activity hints at a greater involvement of cognitive processes, likely because this task is less restricted in terms of movement.

These results confirm that information contained in spectral coherence of source signals, particularly in the gamma band, is a robust marker of motivational state in Parkinsonian subjects.

Discussion

The results of our study show a different physiological pattern depending on the motivation state in a patient with Parkinson's disease. We used a task divided into two motor types (Stop-In and Punch-Through), where we subtly introduced a social hierarchy and decision-making autonomy. This allowed us to see clear changes in EEG activity in important frequency bands, mainly in the beta (15-32 Hz) and gamma (40-80 Hz) bands. Remarkably, these effects were found without any external reward, showing that the Parkinson's brain can respond to implicit motivational signals as the healthy patients. These findings suggest that, even with damage to the dopamine pathways typical of the disease [9] [10], some brain networks stay functional and keep the ability to respond to implicit motivation signals [8].

Activation of Brain Areas in Response to Social Pressure

The supervised classifiers, especially the MLR model, did really well, hitting about 80 - 90% accuracy in the Punch-Through condition (beta and gamma band). This suggests that different motivation levels trigger clear brain signals that we can identify. These findings match what we've seen in studies with healthy people [4], supporting the idea that social pressure, even without real rewards, can light up the areas of the brain that deal with action and planning.

The strong involvement of the beta and gamma band, typically associated with high-level processes like attention coordination, decision-making, and intentional action, suggests that cognitive mechanisms remain at least partially active in the Parkinsonian brain [24] [27].

Moreover, these electrophysiological changes are observed in other neurocognitive disorders such as Alzheimer's disease, where a decline in cortical coherence especially in alpha and beta bands marks a breakdown in network connectivity [25] [26].

We looked closely at one case, but we set things up so that our method could work for other cases too. The main goal here was to create a solid pipeline to analyze both unhealthy and healthy data.

This challenges a too rigid view of motivation deficits in Parkinson's disease and opens new paths for therapeutic approaches based on implicit forms of engagement, especially social ones [28]. To really established this perspective a quantitative comparison with healthy controls and a detailed performance analysis of the movement task are still needed.

Study on Motivation Patterns

We found that when people were highly motivated, there was a noticeable boost in gamma-band synchronization, particularly during the Punch-Through condition. This condition, offering more freedom of action and less preparation, subtly introduced both autonomy and an implicit social hierarchy. This appears to have modulated effort based on the social context, favoring a stronger activation of brain networks involved in attention and moto planning. [1] [2]

These results resonate with the findings of Palva & Palva (2007) [12], who emphasized the role of fast frequency bands in information integration and attentional control. Specifically, the beta band is often tied to top-down motor control [5] [13], while the gamma band is involved in transient neural assemblies that support action intention, perception, and working memory [29]. This gives a helpful overview, but it's not a strict goal or rigid interpretation but rather as a conceptual lens to keep it mind when interpreting the data.

In our study, the Punch-Through task really digs into how things work because it makes you think more and gives a sense of control. That doesn't mean the Stop-In movement didn't work. It just didn't perform as well in classification. This consistent difference points to the Punch-Through setup being better at showing how we put in effort and manage our movements. Basically, the experiment helps us figure out what makes a good biomarker that is, a measure that shows clear differences across conditions. This also shows how a focused experimental design can help find these markers, even in people with Parkinson's disease [30]. Although synchronization was more spatially restricted than in healthy brains, it remained clearly localized in key regions, such as the right temporal and parietal cortices [31] [32].

We made a step forward in this study by using actual source inversion techniques with MNE-Python, which helped us get a clearer picture of brain activity. Unlike earlier research from Ignasi et al. [4] that used ICA methods, our approach gave us better spatial accuracy.

ICA tends to stay near the electrodes, which makes it hard to pinpoint certain areas of the brain. But with MNE-Python and its brain atlases, we could break the brain down into clearer anatomical regions. This methodological upgrade was a key technical objective of our study, as it addressed the limitations of previous work and helped enhance the interpretability of the EEG signal in a clinical context.

EEG Coherence and Plasticity in Neurodegeneration

In many neurodegenerative diseases such as Alzheimer's, a well-documented drop in EEG coherence especially in slow bands like delta and theta is seen as a marker of functional disintegration in cortical networks [33]. It is also important to consider the role of abnormal beta rhythm. In this context, the beta band desynchronization may actively contribute to the disruption of network coordination, highlighting its potential role in the pathophysiology of neurodegeneration [34].

We found something interesting in our research on Parkinson's disease. Typically, you'd see low slow-band coherence with neurodegenerative diseases. But we noticed that the faster frequency bands were actually more in synchronization. In some areas, the gamma band even showed better coherence. This suggests that, unlike in Alzheimer's, where beta-band desynchronization might lead to network issues, the fast-frequency activities in Parkinson's could still be doing their job or finding a way to compensate, hinting at different ways the network might stay coordinated.

This shows that there's still some plasticity in the brain [35]. The right temporal and parietal areas stand out, as they seem to play a key role in this process and might indicate some compensatory actions. These results support current ideas about how the brain reorganizes itself during neurodegeneration [36], suggesting that even in a brain that's declining, motivation can still tap into some functional resources. This not only helps us understand brain activity in Parkinson's disease better but also opens up potential treatment options that focus on this remaining plasticity.

Limitations and Perspectives

We can mention some of the limitations in our study. First, this is a single-case study, which restricts the generalizability of the findings because in our study there is just one patient cause take so much time to clean the data.

The second issue is about how we trained the classifier. We trained the model for an average number of epochs for each condition, which could overlook some of the trial-by-trial differences. This makes the timing less detailed and might hide some subtle motivational effects. In future research, using classifiers on individual epochs could help us analyze motivational changes at the trial level. This is an important step for providing tailored clinical follow-up. This kind of approach adds a lot of strain on the computer since there's more data and the models are more complex. What we can do is use parallel processing and powerful computing clusters. This way, we can train and test on a large scale without losing performance, which makes single-trial decoding possible and ready for clinical use.

EEG does a great job with tracking time, but it struggles a bit when it comes to pinpointing exactly where in the brain things are happening. That's where the source reconstruction using MNE-Python really steps up. Unlike ICA, which breaks down functions but doesn't give a clear picture of where stuff is located, our method uses anatomical atlases (aparc2009) to make brain activity easier to understand.

On top of that, we're starting to test this method on healthy people to see how motivational signals change or stay the same in Parkinson's disease. We're still working through that data. At the same time, we're looking at clinical data to understand how motivation varies in real-life situations, which gives us more insight into how our method can be used.

Finally it'd be great to use smarter ways to analyze connectivity, like Granger Causality [37] or Transfer Entropy [38], to really get what's happening in brain interactions [39]. It would also help to include more patients with different backgrounds and do longer studies to keep an eye on motivational markers as time goes on. Combining EEG with imaging modalities such as fMRI would provide richer spatio-temporal information [40]. Moreover, non-invasive brain stimulation methods like TMS [41] or tDCS [42], targeting regions activated in high-motivation states, might offer novel therapeutic strategies to enhance patient engagement.

In sum, this study introduces a technically refined, replicable pipeline that not only reveals how the Parkinsonian brain can still respond to implicit motivation but also opens promising avenues for future clinical and translational research.

Conclusion

This study explored how implicit motivational signals, especially social ones, influence brain activity in a patient with Parkinson's disease. By adapting a protocol previously used in healthy participants [4], we observed distinct EEG patterns in the beta and gamma bands, particularly during high-motivation conditions like the Punch-Through task. These effects suggest that, despite dopaminergic degeneration, some cortical networks remain responsive and functionally plastic.

This work presents a new way to analyze EEG data with MNE-Python. Instead of the older methods that relied on ICA and had issues with spatial resolution, this approach combines source reconstruction with atlas-based parcellation. This allows for a better look at brain connectivity and classification. Plus, it achieved high accuracy for identifying motivational states, getting as high as 90% in some cases.

The study is still ongoing, and a key next step is to apply this updated analysis pipeline to a broader range of participants. Initially designed for healthy individuals, the pipeline will now be extended to larger cohorts that include both healthy controls and Parkinson's patients. This expansion will allow us to validate and refine our findings across different populations, helping to account for variability between individuals and disease stages.

By comparing these groups, we aim to gain deeper insights into how motivational brain mechanisms evolve in neurodegenerative conditions. Although this work started with a single-case study, these ongoing efforts provide a solid foundation for identifying reliable neural markers of motivation and developing targeted therapeutic approaches to support cognitive and motor engagement in Parkinson's disease and beyond.

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Appendix

Appendix 1 :

Github's link with the entire project of the EEG's analysis :

https://github.com/Ilian10/Salmi_Ilian_Project_M2_IA_Health_BraiNets_2025

On this link, you can find different notebooks for each part of the study. We choose to cut in 5 parts the code to be more easy to understand. If you want to try to run the code you need to concatenate the 5 parts in one notebook in this orders :

- 'Download_'fsaverage'_data_from_mne.ipynb
- Preprocessing sensors's space.ipynb
- Decoding in sensors's space.ipynb
- Preprocessing in sources's space.ipynb
- Decoding in sources's space.ipynb
-

You can also find six videos of the average signal for each movement and each condition on the cortical's space :

Stop-In :

Solo = solo_SI.webm

Easy = easy_SI.webm

Hard = hard_SI.webm

Punch-Through :

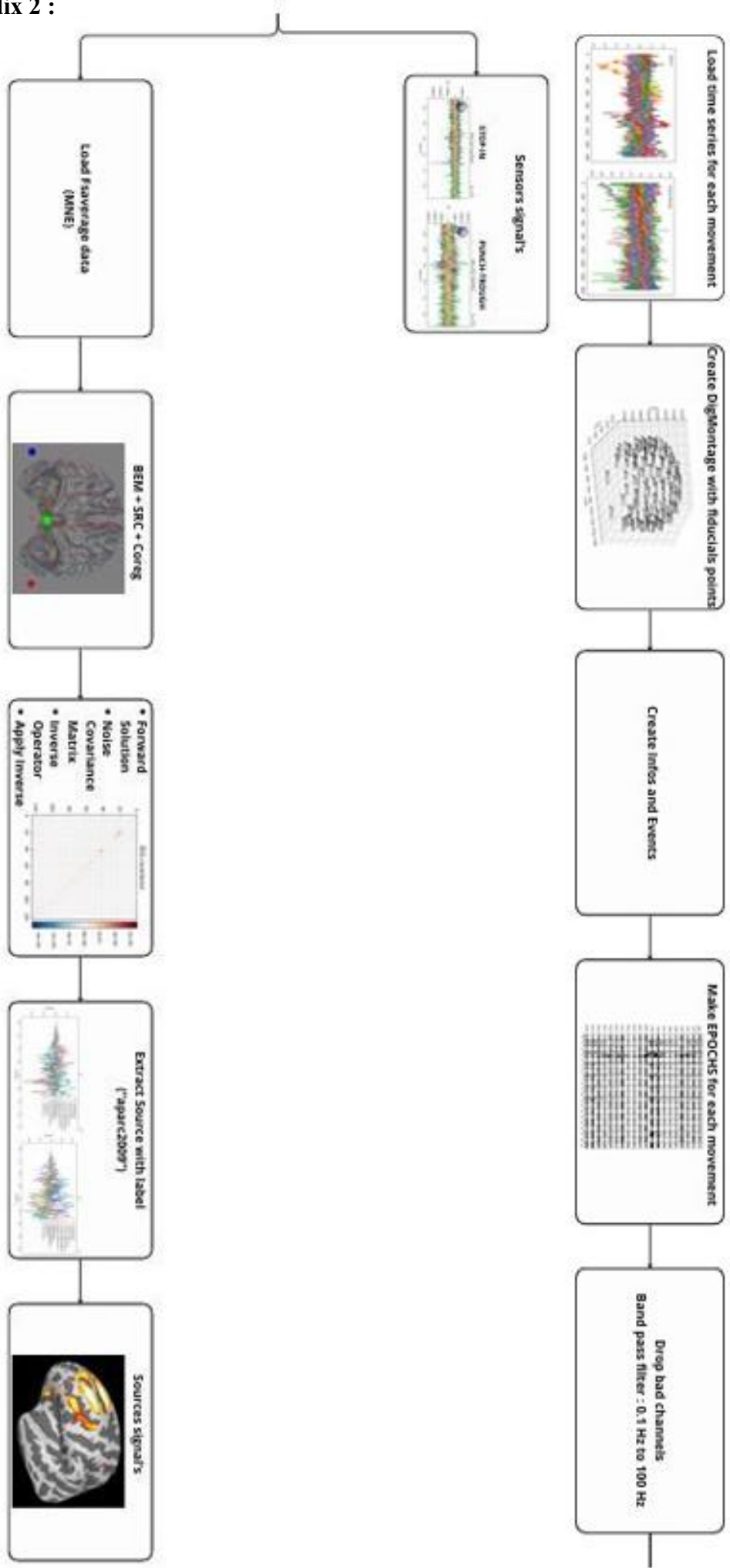
Solo = solo_PT.webm

Easy = easy_PT.webm

Hard = hard_PT.webm

You need to download the video to watch them.

A. Pipeline for preprocessing



Appendix 2 :

B. Pipeline for decoding in sources and sensors's space:

